MSG No. 21.42 C_c222 [Type IV, orthorhombic]

Table 1: Wyckoff site: `4a`, site symmetry: `222`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,4]`
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[5,6,7,8]`
3	$\left[0, 0, \frac{1}{2}\right]$	`[9,10,11,12]`
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[13,14,15,16]`

Table 2: Wyckoff site: `4b`, site symmetry: `222`

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	`[1,2,3,4]`
2	$\left[\frac{1}{2}, 0, 0\right]$	`[5,6,7,8]`
3	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[9,10,11,12]`
4	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[13,14,15,16]`

Table 3: Wyckoff site: `4c`, site symmetry: `2'2'2`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[1,4,10,11]`
2	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[2,3,9,12]`
3	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[5,8,14,15]`
4	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[6,7,13,16]`

Table 4: Wyckoff site: `4d`, site symmetry: `2'2'2`

No.	position	mapping
1	$\left[0, 0, \frac{1}{4}\right]$	`[1,4,10,11]`
2	$\left[0, 0, \frac{3}{4}\right]$	`[2,3,9,12]`
3	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[5,8,14,15]`
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[6,7,13,16]`

Table 5: Wyckoff site: `8e`, site symmetry: `2..`

No.	position	mapping
1	$[x, 0, 0]$	`[1,2]`
2	$[-x, 0, 0]$	`[3,4]`
3	$\left[x + \frac{1}{2}, \frac{1}{2}, 0\right]$	`[5,6]`
4	$\left[\frac{1}{2} - x, \frac{1}{2}, 0\right]$	`[7,8]`
5	$\left[x, 0, \frac{1}{2}\right]$	`[9,10]`
6	$\left[-x, 0, \frac{1}{2}\right]$	`[11,12]`
7	$\left[x + \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[13,14]`

8	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[15,16]`

Table 6: Wyckoff site: `8f`, site symmetry: `2'..`

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{4}\right]$	`[1,10]`
2	$\left[x,\ 0,\ \frac{3}{4}\right]$	`[2,9]`
3	$\left[-x,\ 0,\ \frac{3}{4}\right]$	`[3,12]`
4	$\left[-x,\ 0,\ \frac{1}{4}\right]$	`[4,11]`
5	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	`[5,14]`
6	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	`[6,13]`
7	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[7,16]`
8	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[8,15]`

Table 7: Wyckoff site: `8g`, site symmetry: `.2.`

No.	position	mapping
1	$[0,\ y,\ 0]$	`[1,3]`
2	$[0,\ -y,\ 0]$	`[2,4]`
3	$\left[\frac{1}{2},\ y+\frac{1}{2},\ 0\right]$	`[5,7]`
4	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ 0\right]$	`[6,8]`
5	$\left[0,\ y,\ \frac{1}{2}\right]$	`[9,11]`
6	$\left[0,\ -y,\ \frac{1}{2}\right]$	`[10,12]`
7	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}\right]$	`[13,15]`
8	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	`[14,16]`

Table 8: Wyckoff site: `8h`, site symmetry: `.2'.`

No.	position	mapping
1	$\left[0,\ y,\ \frac{1}{4}\right]$	`[1,11]`
2	$\left[0,\ -y,\ \frac{3}{4}\right]$	`[2,12]`
3	$\left[0,\ y,\ \frac{3}{4}\right]$	`[3,9]`
4	$\left[0,\ -y,\ \frac{1}{4}\right]$	`[4,10]`
5	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{4}\right]$	`[5,15]`
6	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	`[6,16]`
7	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{3}{4}\right]$	`[7,13]`
8	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{4}\right]$	`[8,14]`

Table 9: Wyckoff site: `8i`, site symmetry: `..2`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,4]`
2	$[0,\ 0,\ -z]$	`[2,3]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[5,8]`

continued ...

Table 9

No.	position	mapping
4	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	[6,7]
5	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	[9,12]
6	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	[10,11]
7	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[13,16]
8	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[14,15]

Table 10: Wyckoff site: 8j, site symmetry: ..2

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	[1,4]
2	$\left[0,\ \frac{1}{2},\ -z\right]$	[2,3]
3	$\left[\frac{1}{2},\ 0,\ z\right]$	[5,8]
4	$\left[\frac{1}{2},\ 0,\ -z\right]$	[6,7]
5	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[9,12]
6	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[10,11]
7	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	[13,16]
8	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	[14,15]

Table 11: Wyckoff site: 8k, site symmetry: ..2

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{4},\ z\right]$	[1,8]
2	$\left[\frac{1}{4},\ \frac{3}{4},\ -z\right]$	[2,7]
3	$\left[\frac{3}{4},\ \frac{1}{4},\ -z\right]$	[3,6]
4	$\left[\frac{3}{4},\ \frac{3}{4},\ z\right]$	[4,5]
5	$\left[\frac{1}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	[9,16]
6	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{2}-z\right]$	[10,15]
7	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	[11,14]
8	$\left[\frac{3}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	[12,13]

Table 12: Wyckoff site: 16l, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$[x,\ -y,\ -z]$	[2]
3	$[-x,\ y,\ -z]$	[3]
4	$[-x,\ -y,\ z]$	[4]
5	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z\right]$	[5]
6	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$	[6]
7	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ -z\right]$	[7]
8	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$	[8]
9	$\left[x,\ y,\ z+\frac{1}{2}\right]$	[9]
10	$\left[x,\ -y,\ \frac{1}{2}-z\right]$	[10]

continued ...

Table 12

No.	position	mapping
11	$\left[-x,\, y,\, \frac{1}{2}-z\right]$	[11]
12	$\left[-x,\, -y,\, z+\frac{1}{2}\right]$	[12]
13	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[13]
14	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	[14]
15	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	[15]
16	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[16]